

Xiaona Zhou

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Research Interests

Trustworthy and efficient AI systems; vision-language reasoning; time-series anomaly detection; coding-agent safety and red-teaming; dataset selection for reliable data sharing.

Education

University of Illinois Urbana-Champaign

Ph.D. in Computer Science GPA: 3.96/4.0

Advisor: Dr. Ismini Lourentzou

Champaign, IL

August 2024 - August 2027

Virginia Tech

Master's in Computer Science & Applications, GPA: 3.91/4.0

Advisor: Dr. Ismini Lourentzou

Blacksburg, VA

August 2022-May 2024

Georgia Institute of Technology

Master's in Analytics (Incomplete), GPA: 4.0/4.0

Online Program

August 2021-December 2021

New York City College of Technology (NYCCT), The City University of New York

B.S. Applied Mathematics, GPA: 4.0/4.0, [Salutatorian](#)

Brooklyn, NY

June 2017 - June 2021

Selected Publications

- [1] **Xiaona Zhou**, Ismini Lourentzou, Constantin Brif. IDEAS: Adapting Univariate Anomaly detectors to Multivariate Time Series, **AAAI** (under review), **2027**
- [2] **Xiaona Zhou**, Muntasir Wahed, Tianjiao Yu, Constantin Brif, Ismini Lourentzou. Tiny but Trusted: Efficient Vision-Language Reasoning for Time-Series Anomaly Detection, **NeurIPS** (under review), **2026**
- [3] **Xiaona Zhou**, Yingyan Zeng, Ran Jin, Ismini Lourentzou. Hierarchical Dataset Selection for High-Quality Data Sharing, **AAAI**  Oral, **2026**
- [4] **Xiaona Zhou**, Constantin Brif, and Ismini Lourentzou. mTSBench: Benchmarking Multivariate Time Series Anomaly Detection and Model Selection at Scale, **TMLR**, **2026**
- [5] Muntasir Wahed, **Xiaona Zhou**, Kiet A. Nguyen, Tianjiao Yu, Nirav Diwan, Gang Wang, Dilek Hakkani-Tür, Ismini Lourentzou. Are Code Language Models Robust Against Multi-Turn Malicious Coding Prompts? **EMNLP**, **2025**
- [6] Jiawei Liu, Nirav Diwan, Zhe Wang, Haoyu Zhai, **Xiaona Zhou**, Kiet A. Nguyen, Tianjiao Yu, Muntasir Wahed, Yinlin Deng, Hadjer Benkraouda, Yuxiang Wei, Lingming Zhang, Ismini Lourentzou, Gang Wang. PurpCode: Reasoning for Safer Code Generation. **NeurIPS**, **2025**.
- [7] Yingyan Zeng, **Xiaona Zhou**, Premith Chilukuri, Ismini Lourentzou, Ran Jin. High-Quality Dataset-Sharing and Trade Based on A Performance-Oriented Directed Graph Neural Network, **IEEE Transactions on Automation Science and Engineering (T-ASE)**, **2025**
- [8] Muntasir Wahed, **Xiaona Zhou**, Tianjiao Yu, Ismini Lourentzou. Fine-Grained Alignment for Cross-Modal Recipe Retrieval, **WACV**, **2024**

Research Experience(Academic & Industry)

Graduate Research Assistant, University of Illinois Urbana-Champaign

August 2024 - Present

Advisor: Professor Ismini Lourentzou

Parameter-Efficient VLM for Grounded Time-Series Anomaly Reasoning

- Developing compact VLMs with variable-length continuous latent reasoning, variable-cardinality anomaly localization, and region-grounded explanations..
- Designing a structured-verifier GFlowNet objective to preserve diverse high-quality reasoning trajectories while balancing localization accuracy, explanation faithfulness, and reasoning cost.

Amazon Trusted Coding Agents and Red-Teaming

- Designing multi-model vulnerability-detection oracles and execution-based verifiers to improve vulnerability coverage while reducing false positives in coding-agent-modified web applications.
- Worked on designing a 13-category malicious-code taxonomy and multi-turn code-decomposition attacks to evaluate code LLM robustness.

Vision-Language Time-Series Anomaly Reasoning

- Curated the first explanation-augmented benchmark for vision-language time-series anomaly reasoning, with reward-selected, visually grounded explanations.
- Developed a **family of parameter-efficient VLMs** for anomaly localization and grounded reasoning, outperforming 15 baselines, including large VLMs and **GPT-based** methods, by at least 21 pp in precision and F1.

Multivariate Time-Series Anomaly Detection Benchmarking

- Built the largest benchmark to date for multivariate time-series anomaly detection, comprising 344 labeled time series across 19 datasets from a wide range of application domains.
- Benchmarked 24 detectors under 13 metrics and evaluated unsupervised model-selection methods, finding that

neither detectors nor selectors generalize reliably across datasets.

Dataset Selection for Heterogeneous Training

- Developed a hierarchical Bayesian dataset-selection method that models dataset- and group-level utility to select datasets from heterogeneous pools under resource constraints.
- Improved accuracy by **up to 26.2% on two domain adaptation benchmarks** over state-of-the-art data-selection baselines, with fewer exploration steps and robustness to noisy grouping..

Year-round Research Intern, Sandia National Labs

May 2023 - present

Mentor: Dr. Constantin Brif, Principal Member of Technical Staff, Sandia National Labs

- Developed a detector-agnostic framework that adapts anomaly detectors to multivariate time series with channel-level localization, consistently improving performance over counterparts.
- Built GUI, CLI, and SLURM/HPC workflows for anomaly detection on satellite state-of-health telemetry, enabling reproducible large-scale evaluation and analyst-facing inspection.
- Curated open-source and synthetic time-series benchmarks for evaluating detectors and model-selectors.
- Reimplemented and adapted state-of-the-art time-series anomaly detection methods for internal use.
- Parallelized anomaly-detection workflows, reducing runtime from multiple days to under one hour.

Graduate Research Assistant, Virginia Tech

Spring 2023 - May 2024

Advisor: Professor Ismini Lourentzou, Department of Computer Science, Virginia Tech

- Worked on developing a task-driven, privacy-preserving data-sharing framework for the industrial internet that enabled data-sharing while minimizing the risk of data leakage.
- Deployed hierarchical attention networks to capture participant relationships, assuming that shared datasets beneficial for the immediate downstream task also hold value for other tasks.

Student Researcher, NYCCT

Fall 2020 - Spring 2021

Mentor: Professor Boyan Kostadinov, Department of Mathematics, NYCCT

- Used circular statistics to design and implement algorithms to compute and visualize drosophila behavioral phases, with results published in a peer-reviewed paper.
- Developed a **Shiny App** that independently implements all data analysis and visualizations.

Budget Analyst Intern, Office of Brooklyn Borough President

September 2019 - October 2020

Supervisor: Candice Julien, Capital Budget Director, Office of the Brooklyn Borough President, Eric L. Adams

- Analyzed large-scale capital funding data with statistical and machine learning methods, building visualizations and a five-year allocation report for Brooklyn Borough President Eric Adams to support borough-level budget planning and capital allocation decisions.

Teaching Experience

Graduate Teaching Assistant, University of Illinois Urbana-Champaign

Spring 2026 - present

- CS/IS/STAT 107 Data Science Discovery, CS 222 Software Design Lab

Math Tutor, Research Foundation of CUNY

February 2019 – May 2023

Supervisor: Professor Satyanand Singh, Department of Mathematics, NYCCT

- Helped college students with homework in courses such as College Algebra, Calculus, Linear Algebra, Differential Equations, Statistics, Discrete Math, and more.
- Provided test review and correction assistance. Answered questions students post on an online platform.

Awards & Research Funding

- **Amazon Nova AI Challenge** – Winning Team Member (\$250K team prize) 2025
- **NSF Student Travel Award AAAI** 2026
- **National Science Foundation (NSF) Scholarship** Fall 2018 - Spring 2021
- **Salutatorian Award** – New York City College of Technology Class of 2021
- **SIAM Award** – For outstanding efforts and accomplishments Academic year 2020–2021
- **Academic Excellence Award** – School of Art and Science, New York City College of Technology June 2021

Academic Service and Other Activities

- **Program Committee Member**, AAAI Conference on Artificial Intelligence 2027
- **Reviewer**, ICML (Gold Reviewer, 2026); TMLR; IEEE BigData; PETRA 2023–2026
- **President**, SIAM (Society for Industrial and Applied Mathematics) Student Chapter 2019 – 2021
- **Student Researcher**, Microsoft Data Science Summer School 2020

Relevant Coursework

LLM-Agents, Embodied Artificial Intelligence, Multi-modal Vision, Advanced Machine Learning, Deep Learning Theory, Data Analytics, Software Engineering.

Skills

Computer/Programming Skills Python, R, SQL, C++, Java, MATLAB, Maple, Bash, Git, Linux, $\text{\LaTeX}$, R Markdown.

Libraries and Tools PyTorch, Hugging Face Transformers, PEFT (LoRA), TensorFlow/Keras, scikit-learn, NumPy, pandas, tidyverse, Matplotlib, Seaborn, Shiny, Tableau.